

Arithmetic Sequences and Series



Arithmetic sequences

- 1 Consider the sequence 7, 11, 15, 19, ...
 - a State the common difference.
 - b Write a formula for the n th term.
 - c Find a_{20} .

 - 2 In an arithmetic sequence, $a_3 = 10$ and $a_8 = 25$.
 - a Find the common difference and the first term.
 - b Write a formula for a_n .
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Arithmetic series

- 3 Find the sum of the first 30 terms of the series $5 + 9 + 13 + \dots$

- 4 Consider the multiples of 6 strictly between 100 and 500.
 - a How many are there?
 - b Find their sum.

- 5 A theatre has 18 seats in the first row, and each subsequent row has 3 more seats than the row before. How many seats are there in total in 25 rows?